MATHEMATICS / INDEPENDENT STUDY

MM7033

Advanced Algebra

Complete solutions

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Fields and splitting fields

MM7033-U1.01 · E · 15 min
Learning objective: U1.1 Minimal polynomials
Find the minimal polynomial and degree of $\sqrt{11}$ over $\mathbb{Q}$.
$x^2 - 11$ is monic, irreducible by Eisenstein, and vanishes at the element. It is the minimal polynomial, so the degree is two.
Check your argument. Verify all three properties.

MM7033-U1.02 · E · 15 min
Learning objective: U1.1 Splitting fields
Find a splitting field of $x^2 - 3$ over $\mathbb{Q}$.
The field $\mathbb{Q}\left(\sqrt{3}\right)$ contains both roots $\pm\sqrt{3}$ and is generated by them, so it is a splitting field. It has degree two because the polynomial is irreducible.
Check your argument. State the generation condition as well as splitting.

MM7033-U1.03 · E · 15 min
Learning objective: U1.2 Finite-field subfields
List the subfield sizes in $\mathbb{F}_{3^4}$.
The subfields have sizes $3, 9, 81$, each unique. No other degree can occur by the tower law, and the finite-field subfield theorem supplies these three.
Check your argument. Include uniqueness and exhaustiveness.

MM7033-U1.04 · E · 15 min
Learning objective: U1.2 Separability
Show $x^5 - x$ over $\mathbb{F}_5$ has no repeated roots.
The derivative is $5x^4 - 1 = -1$, so its gcd with the polynomial is one. Thus the polynomial is separable. It in fact has all five field elements as roots.
Check your argument. Use the derivative criterion, with the correct characteristic.

MM7033-U1.05 · C · 30 min
Learning objective: U1.1 Splitting-field degree
Find the degree of the splitting field of $x^3 - 3$ over $\mathbb{Q}$.
Eisenstein gives $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ for the real root α. The other roots are $\alpha\zeta, \alpha\zeta^2$, so the splitting field is $\mathbb{Q}(\alpha, \zeta)$. Since $\mathbb{Q}(\alpha)$ is real, it does not contain nonreal ζ, whose quadratic polynomial is $x^2 + x + 1$. The total degree is six.
Check your argument. Justify both steps of the tower.

MM7033-U1.06 · C · 30 min
Learning objective: U1.1 Embeddings
Describe all $\mathbb{Q}$-embeddings of $\mathbb{Q}\left(\sqrt{2}\right)$ into $\mathbb{C}$.
The generator can map only to $\sqrt{2}$ or $-\sqrt{2}$. Each choice extends uniquely to an embedding by the root-extension theorem, with formula $a + b\sqrt{2} \mapsto a \pm b\sqrt{2}$. Hence there are exactly two.
Check your argument. Show the possible images actually define embeddings.

MM7033-U1.07 · C · 30 min
Learning objective: U1.2 Frobenius order
Prove Frobenius on $\mathbb{F}_{p^n}$ has order exactly n.
The nth iterate is $a \mapsto a^{p^n} = a$. If the kth iterate were identity for $0 < k < n$, all p^n elements would solve $x^{p^k} - x = 0$, contradicting the root bound since $p^k < p^n$. Thus the order is exactly n.
Check your argument. Exclude every smaller positive exponent.

MM7033-U1.08 · C · 30 min
Learning objective: U1.2 Derivatives

Prove a polynomial over a field of characteristic p has derivative zero exactly when all nonzero terms have exponents divisible by p.

The derivative is zero precisely when each $ia_i = 0$. If $a_i \neq 0$, the field property makes this equivalent to $i = 0$ in the prime field, meaning $p|i$. The converse follows by differentiating such monomials.

Check your argument. Use uniqueness of polynomial coefficients.

MM7033-U1.09 · C · 30 min
Learning objective: U1.1 Changing the base

Compare the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ and over $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$.

Over $\mathbb{Q}$ the polynomial is $x^2 - 2$. Over the larger field it is $x - \sqrt{2}$. Minimal polynomials and degrees depend on the specified base, even when the element is unchanged.

Check your argument. State both coefficient fields explicitly.

MM7033-U1.10 · A · 50 min
Learning objective: U1.1 Quartic splitting field

Prove the splitting field of $x^4 - 2$ over $\mathbb{Q}$ has degree eight.

Let α be the positive real fourth root of two. Eisenstein at two gives degree four for $\mathbb{Q}(\alpha)/\mathbb{Q}$. The splitting field is $\mathbb{Q}(\alpha, i)$, since adjoining i supplies all four roots and the ratio of $i\alpha$ to α recovers i. The field $\mathbb{Q}(\alpha)$ is real, so $x^2 + 1$ has no root there and adjoining i has degree two. The tower degree is eight.

Check your argument. Prove the named field is generated by the roots and compute the relative quadratic degree.

MM7033-U1.11 · A · 50 min
Learning objective: U1.2 Subfield intersections

Inside $\mathbb{F}_{p^{12}}$, determine the intersection of its subfields of orders p^4 and p^6.

The intersection is a subfield, so its degree over $\mathbb{F}_p$ divides four and six, hence divides two. Both larger subfields contain the unique order-p^2 subfield, because two divides both degrees and uniqueness identifies their copies inside the ambient field. Thus the intersection has degree at least two and therefore exactly two.

Check your argument. Prove both a lower and upper bound on the intersection.

MM7033-U1.12 · A · 50 min
Learning objective: U1.2 Inseparable irreducibles

Prove $x^p - t$ is irreducible and inseparable over $\mathbb{F}_{p(t)}$.

Decisive idea. Use the prime t to prove irreducibility, then the characteristic-p binomial identity to exhibit inseparability.

Finding the approach. Treat the polynomial first over the UFD $\mathbb{F}_{p[t]}$, where the prime element t is available. The passage to its fraction field is a separate step.

Argument. It is primitive and Eisenstein at t: all nonleading coefficients are divisible by t and the constant term is not divisible by t^2. Gauss's lemma gives irreducibility over the fraction field. Its derivative in x is zero. In a splitting field with root α, the characteristic-p identity gives $x^p - t = (x - \alpha)^p$, so its root repeats and it is inseparable.

The leading coefficient is one, so the polynomial is primitive. The zero intermediate coefficients are divisible by t, and the constant $-t$ is divisible by t but not t^2. Eisenstein therefore applies in the polynomial variable x. In characteristic p, every intermediate binomial coefficient in $(x - \alpha)^p$ vanishes. Since $\alpha^p = t$, this expansion shows one root with multiplicity p, not merely that a formal derivative happens to be zero.

A tempting dead end. A zero derivative does not by itself prove irreducibility. Conversely, an irreducible polynomial over a nonperfect field need not have distinct roots.

Hypotheses. The characteristic is the prime p, and t is an indeterminate, not a chosen element of a finite field. The coefficient field $\mathbb{F}_{p(t)}$ need not be perfect.

Check your argument. Explain why the finite-field separability theorem does not apply to the infinite coefficient field.

MM7033-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Degrees inside a finite field

How many elements of $\mathbb{F}_{64}$ have minimal polynomial of degree six over $\mathbb{F}_2$? Justify that every smaller possible degree has been accounted for.

Decisive idea. Replace a minimal-polynomial degree condition by membership in the union of proper subfields.

Finding the approach. Count the elements lying in proper subfields, taking care not to count their intersection twice. Field degree restricts which subfields are possible.

Argument. For $\alpha \in \mathbb{F}_{64}$, the tower law shows $[\mathbb{F}_2(\alpha) : \mathbb{F}_2]$ divides six, so it is one, two, three, or six. Finite-field subfield structure gives unique subfields of orders two, four, and eight. The two-element field is contained in both larger proper subfields. Their intersection has degree dividing both two and three, hence degree one and size two. Thus exactly $4 + 8 - 2 = 10$ elements have degree less than six. The remaining $64 - 10 = 54$ have degree six. Equivalently these form nine Frobenius orbits of size six, corresponding to nine monic irreducible degree-six polynomials.

A tempting dead end. Subtracting the sizes two, four, and eight counts the two-element field repeatedly. Also excluding only the prime field misses elements of degrees two and three.

Hypotheses. The count uses the uniqueness of each finite subfield and the tower divisibility for intermediate fields. It includes zero among the smaller-degree elements.

Check your argument. List all degree divisors, identify the intersection, and justify the subtraction.

2 Unit 2 · Existence and transcendence

MM7033-U2.01 · E · 15 min
Learning objective: U2.1 Chains
Give a partially ordered set with two maximal elements and no greatest element.
Let the elements be a, b, with no comparison between distinct elements. Both are maximal because neither has a strictly larger element. Neither is greatest because it is not above the other.
Check your argument. Distinguish the definitions of maximal and greatest.

MM7033-U2.02 · E · 15 min
Learning objective: U2.1 Proper ideals
Prove an ideal I of a unital ring is proper exactly when $1 \notin I$.
If $1 \in I$, then every $r = r1$ lies in I, so $I = R$. Conversely if $I = R$, one belongs to it. Taking the negation gives the criterion.
Check your argument. Use the ideal absorption property.

MM7033-U2.03 · E · 15 min
Learning objective: U2.2 Independence
Is $\{t, t^2\}$ algebraically independent over K in $K(t)$?
No. The nonzero polynomial $Y - X^2 \in K[X, Y]$ evaluates to zero at (t, t^2). The singleton $\{t\}$ is independent by the definition of the indeterminate.
Check your argument. Exhibit a nonzero formal polynomial relation.

MM7033-U2.04 · E · 15 min
Learning objective: U2.2 Transcendence bases
Give a transcendence basis of an algebraic extension L/K.
The empty set is a transcendence basis: $K(\varnothing) = K$, and L/K is algebraic by hypothesis. Independence of the empty set means a nonzero constant cannot vanish.
Check your argument. Check both parts of the definition.

MM7033-U2.05 · C · 30 min
Learning objective: U2.1 Chain unions
Prove the union of a nonempty chain of proper ideals is proper and an ideal.
For elements in the union, their two ideals are comparable, so both lie in the larger one; their difference belongs there. Multiplication by a ring element preserves whichever ideal contains the element. If one lay in the union, some member would contain one and be the whole ring, a contradiction.
Check your argument. The empty chain requires a separately chosen upper bound in a Zorn application.

MM7033-U2.06 · C · 30 min
Learning objective: U2.1 Maximal ideals over an ideal
Use maximal-ideal existence in a quotient to show every proper ideal I lies in a maximal ideal.
R/I is a nonzero commutative unital ring, so it has a maximal ideal M'. Its inverse image M contains I, and $R/M \cong (R/I)/M'$ is a field. Therefore M is maximal.
Check your argument. Use the quotient criterion for maximality.

MM7033-U2.07 · C · 30 min
Learning objective: U2.2 Algebraic closure
Show an algebraically closed field has no proper algebraic field extension.
For algebraic α over K, its minimal polynomial splits over the algebraically closed K. Since it is irreducible, it must be linear, so $\alpha \in K$. Every element of an algebraic extension therefore already belongs to K.
Check your argument. Apply the reasoning to every extension element.

MM7033-U2.08 · C · 30 min
Learning objective: U2.2 Transitivity

Prove algebraicity is transitive through a tower, allowing infinite extensions.

If α is algebraic over algebraic L/K, collect the finitely many coefficients of an annihilating polynomial into $F = K(c_1, ..., c_r)$. Then F/K and $F(\alpha)/F$ are finite. The tower law makes $F(\alpha)/K$ finite, so α is algebraic over K.

Check your argument. Do not assume L/K itself has finite degree.

MM7033-U2.09 · C · 30 min
Learning objective: U2.1 Extending independent sets

Use Zorn's lemma to extend a linearly independent subset of a vector space to a basis.

Order the linearly independent subsets containing the prescribed independent set S by inclusion. This poset is nonempty because it contains S. Every chain union remains independent by finite dependence and is an upper bound. Zorn gives a maximal independent superset. If it failed to span, adding a vector outside its span would preserve independence, contradicting maximality. Thus it is a basis.

Check your argument. Preserve the originally specified independent set throughout.

MM7033-U2.10 · A · 50 min
Learning objective: U2.2 A nontrivial transcendence basis

Show $\{t^2\}$ is a transcendence basis of $K(t)/K$.

If a nonzero polynomial P vanished at t^2, then $P(X^2)$ would be a nonzero polynomial vanishing at the indeterminate t, impossible. Thus t^2 is transcendental. The element t is algebraic over $K(t^2)$ through $X^2 - t^2$, so $K(t)/K(t^2)$ is algebraic. These two facts prove the basis property, in every characteristic.

Check your argument. Do not assume separability in characteristic two.

MM7033-U2.11 · A · 50 min
Learning objective: U2.1 Units detected by maximal ideals

Prove an element of a nonzero commutative unital ring is a unit exactly when it belongs to no maximal ideal.

A unit cannot lie in a proper ideal, hence in no maximal ideal. Conversely if r is not a unit, (r) is proper: an equation $1 = sr$ would make it a unit. Maximal-ideal existence places (r) inside a maximal ideal, which contains r. Taking the contrapositive gives the result.

Check your argument. Use commutativity for the principal-ideal/unit equivalence.

MM7033-U2.12 · A · 50 min
Learning objective: U2.2 Maximal independence

Prove every maximal algebraically independent subset B of L/K is a transcendence basis.

Decisive idea. Turn failure of algebraicity over the generated field into a strictly larger independent set.

Finding the approach. Maximality can be used only after showing that adjoining a purported transcendental element preserves independence. Establish that implication with a finite polynomial relation.

Argument. Suppose α is transcendental over $K(B)$. A nonzero polynomial relation among finitely many elements of B and α can be grouped by powers of α. At least one coefficient polynomial in the distinct elements of B is nonzero after evaluation, by independence. Thus the relation makes α algebraic over $K(B)$, impossible. Hence $B \cup \{\alpha\}$ would be independent, contradicting maximality. Every element is therefore algebraic over $K(B)$.

Write a possible relation as $\sum_{j=0}^{n} P_{j(b_1,...,b_r)}\alpha^j = 0$, with not all coefficient polynomials P_j zero. Independence of the distinct b_i means a nonzero P_j cannot evaluate to zero. Thus the displayed relation would be a nonzero polynomial over $K(B)$ vanishing at α, contradicting transcendence. This finite-support argument is valid even when B itself is infinite.

A tempting dead end. A maximal independent set need not contain every element of the extension. It must make the remaining elements algebraic over the field it generates.

Hypotheses. The set is maximal under inclusion among algebraically independent sets. The proof uses finite polynomial expressions and does not require a finite transcendence degree.

Check your argument. Explain why the coefficient polynomial does not vanish identically after evaluation.

MM7033-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Maximal ideals of eventually constant sequences

Let R be the ring of eventually constant real sequences, with coordinatewise operations. Classify its maximal ideals and show that one of them is not finitely generated.

Decisive idea. Use quotient fields to force each coordinate idempotent to zero or one, then exploit finite support to obstruct finite generation.

Finding the approach. The visible evaluation maps give candidates. To prove completeness, examine where the coordinate idempotents can go in an arbitrary field quotient.

Argument. Each coordinate kernel $M_n = \{a : a_n = 0\}$ and the eventual-value kernel I is maximal because the corresponding evaluation onto $\mathbb{R}$ is surjective. The ideal I consists exactly of finite-support sequences. Let M be any maximal ideal. If all e_n belong to M, then $I \subset M$, so maximality of I gives $M = I$. Otherwise choose $e_n \notin M$. In the field R/M its nonzero idempotent image equals one, so $1 - e_n \in M$. Every sequence with nth coordinate zero equals its product with $1 - e_n$, hence $M_n \subset M$ and $M = M_n$. Finally finitely many elements of I have support in some finite set. Any R-linear combination retains support inside that set, so cannot produce e_n outside it. Thus I is not finitely generated.

A tempting dead end. An infinite union of coordinate kernels is not the eventual-value kernel, and an arbitrary union of ideals need not be an ideal at all.

Hypotheses. The sequences are eventually constant, ensuring the eventual-value map is defined and its kernel has finite support. The field quotient is used to classify idempotents.

Check your argument. Verify all evaluation maps, exhaust arbitrary maximal ideals, and prove the support obstruction.

3 Unit 3 · Modules and quotient constructions

MM7033-U3.01 · E · 15 min
Learning objective: U3.1 Annihilators
Compute the annihilator of $[4]$ in the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$.
The annihilator is $3\mathbb{Z}$, because $12|4r$ iff $3|r$. The cyclic submodule generated by $[4]$ is consequently isomorphic to $\mathbb{Z}/3\mathbb{Z}$.
Check your argument. Specify the element annihilator rather than the whole-module annihilator.

MM7033-U3.02 · E · 15 min
Learning objective: U3.1 Module maps
Describe all $\mathbb{Z}$-module maps $\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$.
For each $a \in \mathbb{Z}/5\mathbb{Z}$, the map is $n \mapsto na$. These are additive and hence $\mathbb{Z}$-linear, and every map has this form by additivity. There are five.
Check your argument. Explain existence and exhaustiveness.

MM7033-U3.03 · E · 15 min
Learning objective: U3.2 Submodules
Show an ideal of R is exactly a submodule of R as an R-module.
A submodule of the regular module is an additive subgroup with ra in it whenever $r \in R$ and a belongs. In a commutative ring this is exactly an ideal's absorption condition. Thus the notions coincide.
Check your argument. State the role of commutativity if discussing two-sided ideals.

MM7033-U3.04 · E · 15 min
Learning objective: U3.2 Quotients
Describe the quotient $\frac{\mathbb{Z}^2}{N}$ for $N = \{(a, 0) : a \in \mathbb{Z}\}$.
The map $(a, b) \mapsto b$ is a surjective module homomorphism with kernel N. Thus the quotient is isomorphic to $\mathbb{Z}$, by the first module isomorphism theorem.
Check your argument. Provide the map rather than an unsupported dimension argument.

MM7033-U3.05 · C · 30 min
Learning objective: U3.1 Cyclic homomorphisms
Compute the number of $\mathbb{Z}$-module maps $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$.
A map is uniquely determined by $a = f([1])$ and is well defined exactly when $ma = 0$ modulo n. Writing $d = \gcd(m, n)$, this means a is a multiple of $\frac{n}{d}$ modulo n. There are exactly d such residues, hence d homomorphisms.
Check your argument. Prove the condition is sufficient as well as necessary.

MM7033-U3.06 · C · 30 min
Learning objective: U3.1 Operator modules
Let T on K^2 satisfy $T(e_1) = 0, T(e_2) = e_1$. Show its $K[x]$-module is cyclic and identify it as a quotient.
The vectors e_2 and $xe_2 = e_1$ span, so the module is cyclic. For $f = a_0 + a_1x + x^2q$, $f(T)e_2 = a_0e_2 + a_1e_1$, which vanishes exactly when $a_0 = a_1 = 0$. Thus the annihilator is (x^2) and the module is $K[x]/(x^2)$.
Check your argument. Compute the full annihilator, not only one annihilating polynomial.

MM7033-U3.07 · C · 30 min
Learning objective: U3.2 Isomorphism theorem
In $\mathbb{Z}$, take submodules $A = 4\mathbb{Z}$ and $B = 6\mathbb{Z}$. Identify $(A + B)/B$ and $A/(A \cap B)$.
The two modules are $2\mathbb{Z}/6\mathbb{Z}$ and $4\mathbb{Z}/12\mathbb{Z}$, each cyclic of order three. The isomorphism sends the class of an element of $4\mathbb{Z}$ modulo $12\mathbb{Z}$ to its class modulo $6\mathbb{Z}$. Its kernel and image are exactly those in the second isomorphism theorem.
Check your argument. State the induced map and distinguish the ambient quotients.

MM7033-U3.08 · C · 30 min
Learning objective: U3.2 Simple modules
Classify the simple $\mathbb{Z}$-modules up to isomorphism.
Every simple module over a commutative ring is R/M for a maximal ideal. The maximal ideals of $\mathbb{Z}$ are $p\mathbb{Z}$ with p prime. Thus the simple $\mathbb{Z}$-modules are exactly $\mathbb{Z}/p\mathbb{Z}$. Different primes give different cardinalities.
Check your argument. Exclude the zero module by the definition of simple.

MM7033-U3.09 · C · 30 min
Learning objective: U3.1 Torsion
For a domain R, prove the torsion elements of an R-module form a submodule.
If $am = 0$ and $bn = 0$ with nonzero a, b, then $ab(m + n) = 0$. Since R is a domain, $ab \neq 0$. Also $a(rm) = r(am) = 0$ and $a(-m) = 0$. Zero is torsion, so these elements form a submodule.
Check your argument. The domain hypothesis ensures the product annihilator is nonzero.

MM7033-U3.10 · A · 50 min
Learning objective: U3.1 Maps from a quotient
Prove naturally that $\mathrm{Hom}_{R(R/I,M)}$ is isomorphic to $\{m \in M : Im = 0\}$ as an R-module.
Evaluation at $[1]$ sends a linear map to an element annihilated by I, since $i[1] = 0$. Conversely, if $Im = 0$, the formula $[r] \mapsto rm$ is representative-independent and linear. The two constructions are inverse and preserve addition and scalars. Postcomposition with a module map commutes with evaluation, giving naturality in M.
Check your argument. Check well-definedness and the naturality statement.

MM7033-U3.11 · A · 50 min
Learning objective: U3.2 Splitting a torsion module
Prove $\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ as modules, and identify the images of a generator.
The map $[a]_{\{12\}} \mapsto ([a]_3, [a]_4)$ is well defined with trivial kernel, because an integer divisible by three and four is divisible by twelve. Both sides have twelve elements, so it is onto; alternatively $4r - 3s$ lifts $([r]_3, [s]_4)$. The generator $[1]$ maps to $([1], [1])$.
Check your argument. Check the lifting formula modulo each denominator.

MM7033-U3.12 · A · 50 min
Learning objective: U3.1 Submodule invariance
For a vector space with operator T, prove its $K[x]$-submodules are exactly its T-invariant vector subspaces.
Decisive idea. Reduce stability under an entire polynomial ring to stability under its scalar field and its one algebra generator.
Finding the approach. The indeterminate x acts as the operator T. All other polynomial actions are built from that one operator and the already available vector-space operations.
Argument. A $K[x]$-submodule is a K-subspace and is closed under multiplication by x, hence under T. Conversely a K-subspace stable under T is stable under every power of T by induction, and under every finite linear combination of these powers. Thus it is stable under all $f(T)$ and is a submodule.
If W is invariant under T, induction gives $T^{j(W)} \subset W$ for every nonnegative integer j, including $j = 0$ because $T^0 = \mathrm{id}$. For $f(x) = \sum_{j=0}^{n} a_j x^j$ and $w \in W$, each $a_j T^{j(w)}$ lies in W and their finite sum does too. This verifies closure under every scalar in $K[x]$. In the reverse direction, constant polynomials supply all K-scalar operations, while the polynomial x supplies T.
A tempting dead end. Closure under a single application of T is insufficient unless vector-space closure is also established. A set of eigenvectors need not itself be closed under addition.
Hypotheses. Use the specified module structure $f \cdot v = f(T)v$. For a different action of $K[x]$, the associated operator could be different.
Check your argument. Prove both implications and explain the polynomial action.

MM7033-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: A single nilpotent chain

On K^3 let $T(e_1) = 0$, $T(e_2) = e_1$, and $T(e_3) = e_2$. Determine all T-invariant vector subspaces, with proof that your list is exhaustive.

Decisive idea. A highest nonzero coordinate generates all lower coordinates under repeated application and subtraction.

Finding the approach. The operator moves every coordinate down one position. A highest nonzero coordinate therefore forces an entire initial segment of the basis into the subspace.

Argument. The zero subspace is invariant. If a nonzero invariant subspace W has a vector $ae_1 + be_2 + ce_3$ with $c \neq 0$, applying T^2 gives ce_1, so $e_1 \in W$. Applying T and subtracting be_1 gives ce_2, so $e_2 \in W$. Subtracting the first two coordinates of the original vector gives ce_3, so $W = K^3$. If no such vector exists but one has $b \neq 0$, the same two-step argument gives $e_1, e_2 \in W$, and W is their span. If every vector has only its first coordinate nonzero, a nonzero one spans Ke_1. Thus the complete list is zero, Ke_1, $Ke_1 + Ke_2$, and K^3, all visibly invariant.

A tempting dead end. Not every line is invariant. A line through e_2 already fails because its image contains e_1.

Hypotheses. Nonzero coefficients can be inverted because K is a field. The operator is one chain; a direct sum of multiple chains would allow additional invariant subspaces.

Check your argument. Prove exhaustiveness as well as invariance, treating all three possible highest coordinates.

4 Unit 4 · Free modules and exact sequences

MM7033-U4.01 · E · 15 min
Learning objective: U4.1 Presentations
Give a one-generator, one-relation presentation of $\mathbb{Z}/9\mathbb{Z}$.
The presentation is $\mathbb{Z} \xrightarrow{\times 9} \mathbb{Z} \to \mathbb{Z}/9\mathbb{Z} \to 0$. Reduction is onto and its kernel is $9\mathbb{Z}$, the image of the first map.
Check your argument. Verify exactness at the middle and right terms.

MM7033-U4.02 · E · 15 min
Learning objective: U4.2 Exactness
Check exactness of $0 \to K \xrightarrow{i} K^2 \xrightarrow{p} K \to 0$, where $i(a) = (a, 0)$ and $p(a, b) = b$.
i is injective, p is onto, and $\ker p = \{(a, 0)\} = \mathrm{im} i$. Thus the sequence is short exact. A section is $b \mapsto (0, b)$.
Check your argument. Check all three requirements.

MM7033-U4.03 · E · 15 min
Learning objective: U4.1 Finite support
Give an element of $\prod_{n\geq 0} \mathbb{Z}$ not lying in $\bigoplus_{n\geq 0} \mathbb{Z}$.
The tuple $(1, 1, 1, \ldots)$ belongs to the product because every coordinate is an integer. Its support is infinite, so it is absent from the direct sum.
Check your argument. State the support condition.

MM7033-U4.04 · E · 15 min
Learning objective: U4.1 Free maps
How is a linear map $R^3 \to M$ determined?
An arbitrary triple $m_1, m_2, m_3 \in M$ gives the unique map $(r_1, r_2, r_3) \mapsto r_1 m_1 + r_2 m_2 + r_3 m_3$. Conversely every linear map has this form.
Check your argument. Explain uniqueness by basis expansion.

MM7033-U4.05 · C · 30 min
Learning objective: U4.2 Nonsplitting
Prove $0 \to \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$ does not split.
If s were a section, then $3s([1]) = s(3[1]) = 0$, so $s([1]) = 0$. Reducing this image modulo three gives zero, contradicting that the composite is the identity on $[1]$.
Check your argument. Use linearity and the section equation.

MM7033-U4.06 · C · 30 min
Learning objective: U4.2 Free quotients
Show a short exact sequence with quotient R^n splits.
Choose b_i mapping to the ith standard basis vector under the surjection. Define $s(r_1, \ldots, r_n) = \sum_i r_i b_i$. This is linear and composing with the surjection returns the input vector. Thus it is a section and the splitting lemma applies.
Check your argument. Construct an actual section.

MM7033-U4.07 · C · 30 min
Learning objective: U4.1 Presentation computation
Identify the $\mathbb{Z}$-module generated by u, v with relations $u - 2v = 0$ and $6v = 0$.
Every element becomes a multiple of v, and the remaining relation is $6v = 0$. The map to $\mathbb{Z}/6\mathbb{Z}$ sending v to $[1]$ and u to $[2]$ respects both relations. The reverse map sends $[1]$ to v, and their composites are identities. Hence the module is $\mathbb{Z}/6\mathbb{Z}$.
Check your argument. Prove no extra hidden relation remains by constructing inverse maps.

MM7033-U4.08 · C · 30 min
Learning objective: U4.2 Direct summands
Prove a direct summand of a projective module is projective.
Write $P = A \oplus B$ with P projective. Given $f : A \to N$ and a surjection $M \to N$, compose f with the projection $P \to A$, lift that map to M using projectivity, then restrict along $A \to P$. The resulting map

lifts f. Thus A is projective.
Check your argument. State the extension and restriction maps.

MM7033-U4.09 · C · 30 min
Learning objective: U4.1 Finite rank
Prove R^2 cannot be isomorphic to R^3 for a nonzero commutative unital ring.
Choose a maximal ideal M. Quotienting the two modules by M times themselves would give an isomorphism $(R/M)^2 \cong (R/M)^3$. Vector-space dimension rules this out.
Check your argument. Explain why the module isomorphism descends to these quotients.

MM7033-U4.10 · A · 50 min
Learning objective: U4.2 Projective cyclic example
Show $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$-module.
If the module were projective, its identity would lift through the surjection from $\mathbb{Z}$, giving a section. Such a section must send $[1]$ to an integer killed by two, hence to zero, contradicting the section condition. Therefore it is not projective.
Check your argument. Use the definition of projectivity with fully specified maps.

MM7033-U4.11 · A · 50 min
Learning objective: U4.2 A projective nonfree module
Let $R = K \times K$ and $e = (1, 0)$. Prove Re is projective but is not free.
The decomposition $R = Re \oplus R(1 - e)$ follows by splitting $(a, b) = (a, 0) + (0, b)$. It makes Re a direct summand of the free module R, hence projective. It is nonzero and is annihilated by the nonzero element $(0, 1)$. A nonzero free module has a basis vector not annihilated by any nonzero scalar, since its corresponding coordinate is that scalar. Thus Re is not free of any rank.
Check your argument. Exclude rank zero and justify the annihilator property of free modules.

MM7033-U4.12 · A · 50 min
Learning objective: U4.2 Splitting from a retraction
Given a short exact sequence and a retraction $r : B \to A$ of its injection i, construct the isomorphism $B \cong A \oplus C$.
Decisive idea. Subtract the part already detected by the retraction; the remaining kernel supplies a canonical complement.
Finding the approach. A retraction extracts the A-coordinate. Subtract that coordinate from a general element to isolate a complementary piece that maps bijectively to C.
Argument. The map $p : \ker r \to C$ is injective because a kernel element is $i(a)$ with $a = ri(a) = 0$. It is onto because any lift b of c can be replaced by $b - i(r(b))$. Let s be its inverse. Then $(a, c) \mapsto i(a) + s(c)$ has inverse $b \mapsto (r(b), p(b))$. Direct substitution verifies both identities.
If $b \in \ker(r)$ and $p(b) = 0$, exactness gives $b = i(a)$ and the retraction identity gives $0 = r(b) = a$, proving injectivity. For a chosen lift of c, subtracting $i(r(b))$ changes neither $p(b)$ nor its target c, because $pi = 0$. It makes the retraction zero because $ri = \mathrm{id}$. The inverse of the restricted bijective module map is itself linear, so s is a section. For the composite on B, the difference $b - i(r(b))$ is the unique element of $\ker(r)$ mapping to $p(b)$, hence equals $s(p(b))$.
A tempting dead end. Choosing an arbitrary preimage of every c does not ensure those choices form a linear section. The restriction to $\ker(r)$ makes the lift unique and forces linearity.
Hypotheses. Exactness identifies $\ker(p)$ with the image of the injection, and $ri = \mathrm{id}_A$ is indispensable. A short exact sequence need not split without a retraction or section.
Check your argument. Prove the restriction is bijective before using its inverse.

MM7033-U4.13 · A · 60 min · Optional challenge
Learning objective: U4 Transfer: Lifting maps from a residue ideal
Let $R = \mathbb{Z}/6\mathbb{Z}$ and $M = 2R$. Given any surjective R-linear map $q : N \to P$ and any R-linear map $f : M \to P$, construct an R-linear $h : M \to N$ with $qh = f$. Is M free over R? Justify your answer.
Decisive idea. Use an idempotent to repair a lift, and its complement to detect why the module cannot be free.
Finding the approach. The generator two has a relation that an arbitrary preimage need not satisfy. The other nonzero residue in M, namely four, is an idempotent. It can repair a chosen preimage while leaving its image unchanged.

Argument. Write $e = [4]$. Then $e^2 = e$, $Re = \{[0], [2], [4]\} = M$, and $em = m$ for every $m \in M$. Choose $n \in N$ with $q(n) = f(e)$. Define $h(re) = ren$. If $re = r'e$, then $(r - r')e = 0$ in R, so $(r - r')en = 0$; thus h is well defined. Its formula respects addition and scalar multiplication. Moreover $qh(re) = ref(e) = rf(e) = f(re)$, since $ef(e) = f(e^2) = f(e)$. This constructs the required lift for every q and f, and proves projectivity. However, the nonzero scalar [3] kills all of M. In any nonzero free R-module, multiplying a basis vector by [3] gives a vector with nonzero coordinate [3]. Since M is nonzero, it cannot be free.

A tempting dead end. Choosing any preimage of f([2]) and extending from the generator ignores the relation $[3][2] = 0$. The preimage need not be killed by three. Nor does having three elements make M a one-dimensional R-vector space: R is not a field.

Hypotheses. Surjectivity supplies the preimage, R-linearity lets the idempotent repair it, and the arithmetic modulo six supplies the nontrivial idempotent and annihilator. The obstruction concerns freeness over R, not over a different coefficient ring.

Check your argument. Check the lift's relation and linearity, and give an annihilator argument that excludes every nonzero free module.

5 Unit 5 · Tensor products

MM7033-U5.01 · E · 15 min
Learning objective: U5.1 Tensor relations
Simplify $(2m) \otimes n + m \otimes (3n)$ over $\mathbb{Z}$.
The expression is $2(m \otimes n) + 3(m \otimes n) = 5(m \otimes n)$ by bilinearity.
Check your argument. Use tensor relations, not an interpretation as ordered pairs.

MM7033-U5.02 · E · 15 min
Learning objective: U5.2 Cyclic tensors
Compute $\mathbb{Z}/8\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$.
The tensor is $\mathbb{Z}/4\mathbb{Z}$, since tensoring the second module with $\mathbb{Z}/8\mathbb{Z}$ gives its quotient by eight times itself, equivalently $\mathbb{Z}/(8\mathbb{Z} + 12\mathbb{Z}) = \mathbb{Z}/4\mathbb{Z}$.
Check your argument. Explain the quotient computation.

MM7033-U5.03 · E · 15 min
Learning objective: U5.1 Unit tensor
Give inverse maps proving $R \otimes_R M \cong M$.
The maps are $r \otimes m \mapsto rm$ and $m \mapsto 1 \otimes m$. Their composites are identities because $1m = m$ and $1 \otimes rm = r \otimes m$ by scalar movement.
Check your argument. Check well-definedness by bilinearity.

MM7033-U5.04 · E · 15 min
Learning objective: U5.2 Tensor annihilation
Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/7\mathbb{Z}$.
The tensor is $\mathbb{Q}/7\mathbb{Q} = 0$, since any rational q is seven times $\frac{q}{7}$.
Check your argument. Distinguish this from the Cartesian product of the two groups.

MM7033-U5.05 · C · 30 min
Learning objective: U5.1 Pure tensors
Show $e_1 \otimes e_1 + e_2 \otimes e_2$ in $K^2 \otimes K^2$ is not a pure tensor.
Write $u = (a, b), v = (c, d)$. Its coefficient matrix is $\begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$, whose determinant is zero. The displayed tensor has coefficient matrix the identity, determinant one. The coefficients are uniquely determined by the tensor basis, so no pure tensor can equal it.
Check your argument. Justify the basis-coordinate comparison.

MM7033-U5.06 · C · 30 min
Learning objective: U5.1 Tensor basis
Prove $R^m \otimes_R R^n$ is free of rank mn.
The tensors $e_i \otimes f_j$ span by bilinearity. For each pair (i, j), the bilinear map $(u, v) \mapsto u_i v_j$ induces a linear functional taking value one on its matching tensor and zero on the others. These functionals prove independence. Thus the displayed mn tensors form a basis.
Check your argument. Prove independence over a ring, not only over a field.

MM7033-U5.07 · C · 30 min
Learning objective: U5.2 Quotient tensors
Compute $R/I \otimes_R R/J$.
The tensor is $(R/J)/I(R/J)$, canonically isomorphic to $R/(I + J)$. The map sends $[a] \otimes [b]$ to $ab + (I + J)$, and its inverse sends $r + (I + J)$ to $[1] \otimes [r]$.
Check your argument. State the canonical maps or justify the quotient identifications.

MM7033-U5.08 · C · 30 min
Learning objective: U5.2 Failure of injectivity
Tensor $0 \to \mathbb{Z} \xrightarrow{\times 4} \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to 0$ with $\mathbb{Z}/2\mathbb{Z}$. Describe every resulting map.
The nonzero portion is $\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z}/2\mathbb{Z} \to 0$. The first map is zero; the quotient map induces the identity on the generator tensors. The sequence remains exact at the last two terms but not injective at the first.
Check your argument. Explain why a leading zero cannot be retained as an exact term.

MM7033-U5.09 · C · 30 min
Learning objective: U5.1 Distributivity

Prove $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$.

Define $(m, n) \otimes p \mapsto (m \otimes p, n \otimes p)$, which is bilinear and hence descends. In the reverse direction send $m \otimes p$ to $(m, 0) \otimes p$ and $n \otimes p$ to $(0, n) \otimes p$, then add. Their composites are identities on pure tensors and hence everywhere.

Check your argument. Verify that the formulas respect the universal properties.

MM7033-U5.10 · A · 50 min
Learning objective: U5.2 Right exactness

For a submodule $K \leq M$, identify the kernel of $M \otimes N \to (M/K) \otimes N$.

The kernel is the image of $K \otimes N$, equivalently the submodule generated by all $k \otimes n$ with $k \in K$. Quotienting by this submodule classifies bilinear maps on $M \times N$ vanishing on $K \times N$, exactly the maps descending to $(M/K) \times N$. Thus this quotient is the target tensor. The source map from $K \otimes N$ may have its own kernel.

Check your argument. Give the universal-property argument and distinguish image from an embedded copy.

MM7033-U5.11 · A · 50 min
Learning objective: U5.1 Tensor–Hom correspondence

Prove $\mathrm{Hom}_{R(M \otimes N, P)} \cong \mathrm{Hom}_{R\left(M, \mathrm{Hom}_{R(N,P)}\right)}$ naturally.

A linear map from the tensor yields a bilinear $b(m, n)$, which curries to a linear map into the pointwise module of linear maps $N \to P$. Conversely evaluate such a curried map on n to obtain a bilinear map, then use the tensor universal property. Currying and evaluation are inverse, and pre- or postcomposition commutes with both formulas, proving naturality.

Check your argument. Check linearity in each variable and explain the module structure on Hom.

MM7033-U5.12 · A · 50 min
Learning objective: U5.2 A torsion tensor vanishes

Prove $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}(\mathbb{Q}/\mathbb{Z})} = 0$.

Decisive idea. Choose compatible annihilation and division witnesses, then transfer the scalar across the tensor sign.

Finding the approach. The first factor is torsion and the second is divisible. Choose the annihilating integer from the first factor before choosing a division witness in the second.

Argument. For a pure tensor $x \otimes y$, choose positive n with $nx = 0$. Writing $y = q + \mathbb{Z}$, take $z = \frac{q}{n} + \mathbb{Z}$ so that $nz = y$. This proves the divisibility used here directly. Then $x \otimes y = x \otimes nz = nx \otimes z = 0$. Every tensor is a finite sum of pure tensors, so the whole tensor product vanishes.

For $x = \frac{a}{b} + \mathbb{Z}$, the positive integer b kills x. If $y = \frac{c}{d} + \mathbb{Z}$, a valid witness for $bz = y$ is $z = \frac{c}{bd} + \mathbb{Z}$. No canonical division operation on $\mathbb{Q}/\mathbb{Z}$ is required; existence of one witness for this tensor is enough. The balancing relation transfers the integer from the second factor to the first, where it annihilates the element. Since pure tensors generate the tensor product, vanishing on them settles every element.

A tempting dead end. Nonzero modules can have zero tensor product. Also division by n on residue classes is not a uniquely defined operation; different witnesses can differ by n-torsion.

Hypotheses. The factors are modules over the integers. The first has elementwise torsion and the second is divisible; the same argument works for any pair with those properties.

Check your argument. Explain both torsion and divisibility and conclude for sums, not only pure tensors.

MM7033-U5.13 · A · 60 min · Optional challenge
Learning objective: U5 Transfer: Relations after changing coefficients

Let M be generated over $\mathbb{Z}$ by u,v with relations $2u + 3v = 0$ and $4u + 9v = 0$. Compute $M \otimes_{\mathbb{Z}} \mathbb{Q}$, $M \otimes_{\mathbb{Z}} \mathbb{F}_2$, and $M \otimes_{\mathbb{Z}} \mathbb{F}_3$. Decide whether M is zero.

Decisive idea. Track relations through three coefficient systems and distinguish information lost over the rationals from information visible modulo primes.

Finding the approach. The presentation is better suited to a change of coefficients than a search for individual elements. Right exactness preserves its cokernel description.

Argument. Use the matrix with columns $(2, 3)$ and $(4, 9)$, whose determinant is six. Over $\mathbb{Q}$ it is invertible, so the cokernel and hence $M \otimes \mathbb{Q}$ are zero. Over $\mathbb{F}_2$ the columns are $(0, 1)$ and $(0, 1)$, so the image is the line spanned by the second basis vector and the cokernel is one-dimensional, generated by u. Over $\mathbb{F}_3$ they are $(2, 0)$ and $(1, 0)$, so the cokernel is one-dimensional, generated by v. In particular M is not zero. An explicit nonzero map $M \to \mathbb{F}_2$ sends u to one and v to zero; both defining relations map to zero, so it is well defined.

A tempting dead end. A zero tensor with the rationals does not imply the original abelian group is zero. It can mean its elements are torsion. Nor may one retain the original matrix rank after reducing coefficients.

Hypotheses. Tensor is right exact, which is enough to compute the cokernel. No preservation of an injective first map is being assumed.

Check your argument. Show the columns and ranks in all three fields, identify the generators, and exhibit a nonzero quotient of M.

6 Unit 6 · Localization

MM7033-U6.01 · E · 15 min
Learning objective: U6.1 Localized integers
Describe $\mathbb{Z}\left[\frac{1}{3}\right]$ as a subring of $\mathbb{Q}$.
It is $\{\frac{a}{3^n} : a \in \mathbb{Z}, n \geq 0\}$. Addition and multiplication keep this form. It is the smallest subring of $\mathbb{Q}$ containing $\mathbb{Z}$ and $\frac{1}{3}$.
Check your argument. Specify allowed denominators.

MM7033-U6.02 · E · 15 min
Learning objective: U6.1 Units at a prime
Which of $2, 5, 10$ are units in $\mathbb{Z}_{(5)}$?
Two is a unit, with inverse $\frac{1}{2}$. Five and ten lie in the unique maximal ideal and are not units. If either had an inverse with denominator not divisible by five, cross multiplication would force that denominator divisible by five.
Check your argument. Justify the nonunit claims.

MM7033-U6.03 · E · 15 min
Learning objective: U6.1 Vanishing
In $\mathbb{Z}/10\mathbb{Z}$, localize at powers of two. Show the class of five becomes zero.
The element two belongs to the denominator set and annihilates five. Hence $\frac{[5]}{1} = 0$ in the localization by the defining fraction relation.
Check your argument. Do not cancel two inside the original ring.

MM7033-U6.04 · E · 15 min
Learning objective: U6.2 Localizing a cyclic module
Compute $\mathbb{Z}\left[\frac{1}{2}\right] \otimes_{\mathbb{Z}} \mathbb{Z}/8\mathbb{Z}$.
The tensor is $\mathbb{Z}\left[\frac{1}{2}\right]/8\mathbb{Z}\left[\frac{1}{2}\right] = 0$, since multiplication by eight is invertible in the localized ring.
Check your argument. Explain why the quotient is zero.

MM7033-U6.05 · C · 30 min
Learning objective: U6.1 Injectivity criterion
Prove the map $R \to S^{-1}R$ is injective exactly when no element of S annihilates a nonzero element of R.
The kernel is $\{a : ua = 0 \text{ for some } u \in S\}$. It is zero precisely under the stated condition. This equivalence also handles the case $0 \in S$, where injectivity fails unless R is zero.
Check your argument. State both directions and the zero-ring edge case.

MM7033-U6.06 · C · 30 min
Learning objective: U6.1 Identifying a localization
Prove localizing $\mathbb{Z}/6\mathbb{Z}$ at powers of two gives $\mathbb{F}_3$.
Reduction gives a map to $\mathbb{F}_3$ sending every power of two to a unit, so it extends to the localization. In that localization three is zero because two kills it. Thus the original map factors through $\mathbb{Z}/3\mathbb{Z}$, and two is already invertible there. The resulting maps in both directions agree on original elements and inverses of two, so they are inverse.
Check your argument. Use the universal property to verify the inverse maps.

MM7033-U6.07 · C · 30 min
Learning objective: U6.2 A common annihilator
If M is finitely generated and $S^{-1}M = 0$, prove some $s \in S$ kills all of M.
For generators $m_1, \ldots, m_n$, the vanishing criterion gives $s_i \in S$ with $s_i m_i = 0$. The product $s = \prod_i s_i$ belongs to S and kills every generator, hence every linear combination. Thus $sM = 0$. For the zero module choose $s = 1$.
Check your argument. Use finite generation explicitly.

MM7033-U6.08 · C · 30 min
Learning objective: U6.2 Quotients and localization

Show $S^{-1}(M/N) \cong S^{-1}M/\operatorname{im}(S^{-1}N)$.

Localizing $N \to M \to M/N \to 0$ remains exact, so the target is the quotient of $S^{-1}M$ by the image of the localized submodule. Explicitly $\frac{m}{s}$ maps to $\frac{m+N}{s}$. Since localization preserves injections, that image can also be identified with $S^{-1}N$.

Check your argument. Explain both the image formulation and the injective identification.

MM7033-U6.09 · C · 30 min
Learning objective: U6.1 Local rings

Show the nonunits in $R_{\mathfrak{p}}$ form its unique maximal ideal.

The numerator-in-$\mathfrak{p}$ fractions form a proper ideal by the prime-localization theorem. Every fraction outside it has an invertible numerator and is a unit. No element of a proper ideal can be a unit. Thus this ideal is exactly the nonunits and contains every proper ideal, making it uniquely maximal.

Check your argument. Use properness as well as the unit calculation.

MM7033-U6.10 · A · 50 min
Learning objective: U6.2 Detecting zero locally

Prove an R-module M is zero if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$. No finite-generation hypothesis is allowed.

If $m \in M$ is nonzero, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. If $\frac{m}{1} = 0$ in that localization, some $s \notin \mathfrak{m}$ kills m, placing s in the annihilator and hence in $\mathfrak{m}$, a contradiction. Thus some localization is nonzero. The contrapositive proves the claim.

Check your argument. Work with one element's annihilator; do not assume a finite generator list.

MM7033-U6.11 · A · 50 min
Learning objective: U6.1 Localization in stages

Show inverting f and then g in a commutative ring is equivalent to inverting fg.

A map out of the iterated localization is exactly a map from R sending both f, g to units. This is equivalent to sending fg to a unit: if u inverts fg, then gu inverts f and fu inverts g, using commutativity. Thus the two rings have the same universal property and are canonically isomorphic.

Check your argument. Prove the implication from the product being a unit.

MM7033-U6.12 · A · 50 min
Learning objective: U6.2 Flatness of localization (proof reconstruction)

Prove $S^{-1}R$ is flat by explicitly checking that an injection remains injective after localization and identifying localization with tensoring.

Decisive idea. Transport the zero-fraction witness across the injection and then use the tensor-localization comparison.

Finding the approach. A localized element is zero when some allowed denominator kills its numerator. This is weaker than the numerator being zero and is exactly the right invariant to use.

Argument. For injective $f : M \to N$, if $\frac{f(m)}{s} = 0$, there is $u \in S$ with $f(um) = 0$. Injectivity gives $um = 0$, so $\frac{m}{s} = 0$. Hence localization preserves injections. The natural isomorphism $S^{-1}R \otimes M \cong S^{-1}M$ transfers this property to tensoring with $S^{-1}R$, which is flatness.

The tensor comparison sends $\left(\frac{r}{s}\right) \otimes m$ to $\frac{rm}{s}$. Its inverse sends $\frac{m}{s}$ to $\left(\frac{1}{s}\right) \otimes m$. The fraction relation and the tensor balancing relation make these formulas well defined, and substituting a pure tensor or a fraction checks the two composites. In the injectivity argument, if the image fraction is zero, obtain a witness $u \in S$ before using injectivity of f; the result $um = 0$ is precisely the witness needed in the source localization.

A tempting dead end. Do not cancel a denominator in the original module or infer $m = 0$ from $\frac{m}{s} = 0$. Localization deliberately kills S-torsion.

Hypotheses. The ring is commutative and S is multiplicatively closed. Flatness concerns preservation of every module injection; one numerical example is not sufficient.

Check your argument. State the tensor identification and the exact vanishing criterion.

MM7033-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Two localized views of a crossing

Let $A = K[x,y]/(xy)$. Consider the map $A \to K[x,x^{-1}] \times K[y,y^{-1}]$ obtained by setting y=0 in the first coordinate and x=0 in the second. Is it injective or surjective? Describe its image exactly.

Decisive idea. Find a unique normal form before comparing local images; the shared constant measures the compatibility left at the intersection.

Finding the approach. The relation kills all mixed monomials. A normal form reveals both what each localization sees and the one piece of information the two views must share.

Argument. Every element has a unique expression $c + xf(x) + yg(y)$: discard monomials containing both variables, and uniqueness follows because a polynomial in the ideal (xy) has only mixed monomials. Its image is $(c + xf(x), c + yg(y))$. If both coordinates vanish, all their polynomial coefficients vanish, so the original element is zero and the map is injective. Its image consists exactly of pairs of ordinary polynomials in x and y having the same constant term; the displayed expression constructs a preimage for each such pair. It is not surjective, since $(x^{-1}, 0)$ is not in the image, and even $(1,0)$ violates the equal-constant requirement.

A tempting dead end. Two localized descriptions cannot necessarily be chosen independently. Also localization does not mean every tuple of fractions comes from a single unlocalized element.

Hypotheses. The relation is exactly xy=0 over a field, so the monomial normal form has no additional identifications. Laurent-polynomial coordinates are the localizations where x or y is inverted.

Check your argument. Prove uniqueness of normal form, injectivity, both directions of the image description, and failure of surjectivity.

7 Unit 7 · Multilinear algebra

MM7033-U7.01 · E · 15 min
Learning objective: U7.1 Exterior arithmetic
Expand $(e_1 + 2e_2) \wedge (3e_1 + e_2)$ in $\wedge^2 R^2$.
The expansion is $e_1 \wedge e_2 + 6e_2 \wedge e_1 = -5e_1 \wedge e_2$. The coefficient is interpreted in R, so it can vanish in characteristic five.
Check your argument. Keep the sign and coefficient-ring interpretation.

MM7033-U7.02 · E · 15 min
Learning objective: U7.1 Exterior rank
Give a basis and rank of $\wedge^2 R^4$.
A basis is $e_1 \wedge e_2, e_1 \wedge e_3, e_1 \wedge e_4, e_2 \wedge e_3, e_2 \wedge e_4, e_3 \wedge e_4$. Its rank is $\binom{4}{2} = 6$.
Check your argument. Use increasing indices once each.

MM7033-U7.03 · E · 15 min
Learning objective: U7.2 Trace and determinant
Find trace and determinant of $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ over $\mathbb{Z}$.
The trace is five and determinant six. Equivalently the operator acts on the top wedge by multiplying $e_1 \wedge e_2$ by six.
Check your argument. Connect at least one computation to the intrinsic definition.

MM7033-U7.04 · E · 15 min
Learning objective: U7.1 Repeated inputs
Why is $v \wedge v = 0$ even in characteristic two?
The exterior power is defined by killing tensors with repeated inputs, so $v \wedge v = 0$ in every characteristic. In characteristic two it does not follow merely from the equation $v \wedge v = -v \wedge v$, which would only say twice it is zero.
Check your argument. Distinguish alternating from a sign-only condition.

MM7033-U7.05 · C · 30 min
Learning objective: U7.2 Cyclic trace
Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for square matrices over a commutative ring.
$\operatorname{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji}$. Commutativity of the ring changes each term to $b_{ji} a_{ij}$, and interchanging indices gives $\operatorname{tr}(BA)$. No division or field assumption is required.
Check your argument. Identify where commutativity enters.

MM7033-U7.06 · C · 30 min
Learning objective: U7.2 Similarity invariance
Prove similar matrices have equal determinant and trace.
$\det(P^{-1}AP) = \det(P)^{-1} \det(A) \det(P) = \det(A)$. Also $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A)$. The inverse determinant relation follows by taking determinants of $PP^{-1} = 1$.
Check your argument. Justify the inverse identities.

MM7033-U7.07 · C · 30 min
Learning objective: U7.1 Functorial exterior powers
Show $\wedge^{n(TS)} = (\wedge^n T)(\wedge^n S)$.
On $m_1 \wedge \ldots \wedge m_n$, both sides give $TSm_1 \wedge \ldots \wedge TSm_n$. Such wedges generate the exterior power, and both maps are linear, so equality holds everywhere.
Check your argument. Explain why checking generators suffices.

MM7033-U7.08 · C · 30 min
Learning objective: U7.2 Rank-one endomorphisms
Let $T : M \to M$ have form $T(m) = \varphi(m)v$ on a finite free module. Prove $\operatorname{tr}(T) = \varphi(v)$.
For basis e_i with coordinate functionals ε_i, the trace is $\sum_i \varepsilon_{i(Te_i)} = \sum_i \varphi(e_i)\varepsilon_{i(v)}$. Linearity of φ and $v = \sum_i \varepsilon_{i(v)} e_i$ identify this with $\varphi(v)$.
Check your argument. Show that the result does not depend on the chosen basis.

MM7033-U7.09 · C · 30 min
Learning objective: U7.1 Vanishing top powers

Prove $\wedge^n R^r = 0$ for $n > r$.

Multilinearity expresses each wedge as a linear combination of wedges of standard basis elements. With more slots than available indices, each term repeats an index and is zero by alternation. Hence every generator and thus every element vanishes.

Check your argument. Use the pigeonhole step explicitly.

MM7033-U7.10 · A · 50 min
Learning objective: U7.2 Dual bases

For a finite projective module, prove the trace formula is independent of the chosen dual basis.

For dual families (m_i, φ_i) and (n_j, ψ_j), expand $m_i = \sum_j \psi_{j(m_i)} n_j$. The first trace becomes $\sum_{i,j} \psi_{j(m_i)} \varphi_{i(Tn_j)}$. For fixed j, the reconstruction $Tn_j = \sum_i \varphi_{i(Tn_j)} m_i$ makes the inner sum equal $\psi_{j(Tn_j)}$. Summing gives the second trace formula.

Check your argument. Use only dual-basis identities, not an assumption that either family is linearly independent.

MM7033-U7.11 · A · 50 min
Learning objective: U7.1 Exterior direct-sum decomposition

For finite free modules M, N, prove $\wedge^2 (M \oplus N) \cong \wedge^2 M \oplus (M \otimes N) \oplus \wedge^2 N$.

Choose bases e_i, f_j for the two summands. The exterior basis on their concatenation splits into $e_i \wedge e_k$ with $i < k$, $e_i \wedge f_j$, and $f_j \wedge f_l$ with $j < l$. These match bases of the three displayed modules. The map is induced by their inclusions, with the middle term $m \otimes n \mapsto (m, 0) \wedge (0, n)$. Thus it is independent of the chosen bases, whose role was to prove bijectivity.

Check your argument. Define the canonical map and use the basis decomposition to establish it is an isomorphism.

MM7033-U7.12 · A · 50 min
Learning objective: U7.2 Finite-order operators

Let $T^5 = 1$ on a finite-dimensional vector space. Show $\det(T)^5 = 1$, and give a reason this does not force T to be the identity.

Decisive idea. Separate a scalar invariant's necessary constraint from the much stronger claim that the operator itself is fixed.

Finding the approach. The determinant compresses an operator to one scalar. Apply multiplicativity to the stated relation, then test how much information that scalar has forgotten.

Argument. Multiplicativity gives $\det(T)^5 = \det(T^5) = 1$. Over $\mathbb{C}$, multiplication by a primitive fifth root of unity on a one-dimensional space is nonidentity but its fifth power is identity. Even determinant one would not force a higher-dimensional operator to be identity, for example a diagonal matrix with reciprocal nontrivial eigenvalues.

For a sharper counterexample with determinant one, choose a primitive fifth root ζ over $\mathbb{C}$ and take $T = \operatorname{diag}(\zeta, \zeta^{-1})$. It satisfies $T^5 = \mathrm{id}$ and $\det(T) = 1$, yet T is not identity because its first basis vector is multiplied by $\zeta \neq 1$. Thus even knowing the determinant exactly, rather than merely its fifth power, cannot recover the operator.

A tempting dead end. The determinant is not an injective function on the general linear group. A scalar relation satisfied by the determinant is a necessary consequence, not a complete description of the matrix.

Hypotheses. Finite dimension makes the ordinary determinant available. The complex field supplies the stated nontrivial roots of unity; the example is not claimed over every field.

Check your argument. Separate the scalar determinant constraint from the full operator.

MM7033-U7.13 · A · 60 min · Optional challenge
Learning objective: U7 Transfer: The determinant of a projection

Let $P^2 = P$ on a finite-dimensional K-vector space and let r be its rank. Prove $\det(I + tP) = (1 + t)^r$ in $K[t]$. Does $\operatorname{tr}(P) = 0$ force $P = 0$?

Decisive idea. Split an idempotent into its image and kernel, then read the determinant and trace from their distinct actions.

Finding the approach. The idempotent identity supplies a direct decomposition without any assumptions about eigenvalue splitting. Compute the polynomial invariant on that decomposition.

Argument. Every vector is $Pv + (v - Pv)$ with the first part in the image and the second in the kernel. Their intersection is zero: if $w = Pv$ and $Pw = 0$, then $w = P^2v = Pw = 0$. Thus the space is the direct sum of image and kernel. P is identity on its r-dimensional image and zero on its kernel, so in an adapted basis $I + tP$ is diagonal with r entries $1 + t$ and all remaining entries one. Multiplying gives the formula. The same basis gives trace equal to $r \cdot 1_K$. In characteristic zero, trace zero forces r=0 and P=0. In characteristic p, the identity on K^p has trace zero but is nonzero, so the implication fails.
A tempting dead end. Trace zero is a scalar equation in K; it does not always say the ordinary integer rank is zero. No division by the characteristic is available.
Hypotheses. Finite dimension permits a basis and determinant. The rank identity is valid in every characteristic, but the trace conclusion depends on characteristic.
Check your argument. Prove the direct sum, work over the polynomial ring, and give a positive-characteristic counterexample.

8 Unit 8 · Noetherian methods and applications

MM7033-U8.01 · E · 15 min
Learning objective: U8.1 Noetherian examples
Explain why every finite-dimensional vector space is a Noetherian module over its field.
The dimensions in an increasing chain are nondecreasing integers bounded by the space's finite dimension. They eventually stabilize, and nested subspaces of equal finite dimension are equal. Thus the chain stabilizes.
Check your argument. Justify equality once dimensions stop increasing.

MM7033-U8.02 · E · 15 min
Learning objective: U8.1 Hilbert's theorem
Why is $K[x, y, z]$ Noetherian for a field K?
A field is Noetherian because its ideals are zero and the whole field. Hilbert's theorem successively gives Noetherianity of $K[x]$, then $K[x, y]$, then $K[x, y, z]$.
Check your argument. Name the base case and each finite-variable extension.

MM7033-U8.03 · E · 15 min
Learning objective: U8.2 Coordinate rings
Show $K[x, y]/(xy)$ is not a domain.
The classes of x and y are nonzero: a multiple of xy cannot equal either polynomial, as evaluation at $y = 0$ or $x = 0$ confirms. Their product is zero by the defining relation. Hence the quotient has zero divisors.
Check your argument. Prove the two factors are nonzero.

MM7033-U8.04 · E · 15 min
Learning objective: U8.2 Specialization
Compute $R^3 \otimes_R R/M$ for a maximal ideal M.
The result is $(R/M)^3$, a three-dimensional vector space over the residue field. The standard basis tensors give its basis.
Check your argument. Identify the coefficient field.

MM7033-U8.05 · C · 30 min
Learning objective: U8.1 Quotients
Prove a quotient of a Noetherian module is Noetherian.
An increasing chain of quotient submodules lifts to a chain of submodules containing the kernel. The original module is Noetherian, so the lifted chain stabilizes. Taking images under the quotient map shows the original quotient chain stabilizes.
Check your argument. Use the submodule correspondence.

MM7033-U8.06 · C · 30 min
Learning objective: U8.1 Finitely generated submodules
Prove the ascending chain condition implies every submodule is finitely generated.
A non-finitely-generated submodule allows a choice m_1, then m_2 outside Rm_1, then m_3 outside their span, and so on. The resulting finitely generated spans form a strictly increasing chain, contradicting the ascending chain condition. Thus every submodule is finitely generated.
Check your argument. Explain why the process cannot terminate under the contrary assumption.

MM7033-U8.07 · C · 30 min
Learning objective: U8.1 Infinite polynomial ring
Show $K[x_1, x_2, \ldots]$ is not Noetherian.
The chain $(x_1) \subset (x_1, x_2) \subset \ldots$ is strictly increasing. Indeed setting the first n variables to zero kills the nth ideal but does not kill x_{n+1}. Thus the chain never stabilizes.
Check your argument. Prove strictness at every stage.

MM7033-U8.08 · C · 30 min
Learning objective: U8.2 Local coordinate calculation
Compute the localization at x of $A = K[x, y]/(xy)$.
The localization is $K[x, x^{-1}]$. A map sends the class of x to x, its inverse to x^{-1}, and y to zero. Conversely the invertible class of x defines a map from the Laurent polynomial ring into A_x. The composites fix all generators, including inverses, so are identities.
Check your argument. Give both maps to rule out extra relations.

MM7033-U8.09 · C · 30 min
Learning objective: U8.2 Finite presentation
Prove every finitely generated module over a Noetherian ring is finitely presented.
Choose $R^n \to M$ onto. The finite free module is Noetherian, so its kernel has finitely many generators. A map from R^m taking basis elements to those generators gives an exact presentation $R^m \to R^n \to M \to 0$.
Check your argument. Identify where Noetherianity is used.

MM7033-U8.10 · A · 50 min
Learning objective: U8.1 Localization preserves Noetherianity
Prove a localization of a Noetherian ring is Noetherian.
Let J be an ideal of $S^{-1}R$ and $I = \{a : \frac{a}{1} \in J\}$. Since R is Noetherian, $I = (a_1, ..., a_n)$. If $\frac{a}{s} \in J$, multiplying by $\frac{s}{1}$ gives $\frac{a}{1} \in J$, so $a \in I$. Thus $\frac{a}{s}$ is a combination of $\frac{a_i}{1}$ with coefficients in the localization. Conversely these generators lie in J, so they generate it. Every ideal is finitely generated.
Check your argument. Prove the contraction–extension equality used in the argument.

MM7033-U8.11 · A · 50 min
Learning objective: U8.2 Eigenspace decomposition
Suppose $T^2 = 1$ on a vector space over a field of characteristic not two. Construct projections onto the two eigenspaces and prove a direct-sum decomposition.
Let $P_+ = \frac{1+T}{2}$ and $P_- = \frac{1-T}{2}$. Using $T^2 = 1$ gives $P_+^2 = P_+$, $P_-^2 = P_-$, $P_+P_- = 0$, and $P_+ + P_- = 1$. Also $TP_+ = P_+$ and $TP_- = -P_-$. Hence their images lie in the two eigenspaces and every vector is their sum. The intersection is zero because the projections are orthogonal, giving the direct sum.
Check your argument. The inverse of two is essential; state the characteristic hypothesis.

MM7033-U8.12 · A · 50 min
Learning objective: U8.1 Proof reconstruction: Hilbert reduction step
In the proof of Hilbert's theorem, explain in detail why finitely many polynomials with leading coefficients generating the leading-coefficient ideal suffice together with finitely many bounded-degree polynomials.
Decisive idea. Combine a terminating degree-reduction procedure with a separately finitely generated remainder space.
Finding the approach. Degree reduction needs both a decreasing measure and a base case. The leading-coefficient generators supply the first; the bounded-degree module supplies the second.
Argument. Let f_i have degrees d_i and put $N = \max d_i$. For any ideal element of degree $d \geq N$, express its leading coefficient as $\sum r_i \operatorname{lc}(f_i)$ and subtract $\sum r_i x^{d-d_i} f_i$ to lower the degree. Repeating terminates below N. The ideal's intersection with polynomials of degree less than N is finitely generated as an R-module because R^N is Noetherian. Those generators finish the reduction.
The exponents $d - d_i$ are nonnegative because $d \geq N$. Therefore the subtraction uses actual polynomials and stays in the ideal. Its leading coefficient vanishes by the chosen expression, so either the remainder is zero or its degree is strictly smaller. The natural-number degree cannot decrease forever. Below N, write the remainder as an R-linear combination of the selected g_j; this is also an R[x]-linear combination. Combining it with every subtraction expresses the original polynomial using the finite union of the two lists. When $N = 0$, the only remainder below degree zero is zero.
A tempting dead end. Leading coefficients alone do not let one use negative powers of x to eliminate a low-degree remainder. That would leave the polynomial ring.
Hypotheses. The base ring is Noetherian so the submodule inside the finite free module R^N is finitely generated. This is a proof-reconstruction exercise, with attention to the termination and base-case obligations.

Check your argument. Explain termination and why coefficient generation alone does not handle low degrees.

MM7033-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A subring with too many generators

Inside $K[x, y]$, let $B = K + xK[x, y]$. Prove B is a subring and is not Noetherian, although $K[x, y]$ is Noetherian.

Decisive idea. Compare the allowed coefficient ring with the ambient one; a single coefficient layer detects an infinite ascending chain.

Finding the approach. The unusual subring restricts the multipliers available when generating an ideal. Focus on the coefficient of the first power of x to expose that restriction.

Argument. Sums and products of expressions $c + xf$ again have this form, and one belongs to B, so it is a subring. Let $I_n = (x, xy, \ldots, xy^n)$ as an ideal of B. If xy^{n+1} belonged to I_n, write it as $\sum_{j=0}^{n}(c_j + xf_j)xy^j$. Comparing coefficients of x in $K[y][x]$ gives $y^{n+1} = \sum_{j=0}^{n} c_j y^j$, impossible by degrees. Thus I_n is strictly contained in I_{n+1} for every n, violating the ascending chain condition. Hilbert's basis theorem makes $K[x, y]$ Noetherian because K is a field; it makes no claim that all its subrings inherit this property.

A tempting dead end. In the ambient ring every displayed generator is a multiple of x, but the multiplier y need not lie in B. Ideal generation depends on which ring supplies coefficients.

Hypotheses. The coefficients c_j are in K because B has constant x-degree-zero part. The polynomial coefficient comparison takes place in a domain with independent indeterminates.

Check your argument. Check subring closure and prove every inclusion in the ideal chain is strict.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M1.01 · E · 15 min
Learning objective: U1 Finite fields
List the subfield sizes of $\mathbb{F}_{2^8}$ and justify that the list is complete.
The sizes are $2, 4, 16, 256$, one subfield of each size. Necessity follows from the tower law and sufficiency and uniqueness from the finite-field subfield theorem.
Check your argument. State all three aspects: necessity, existence, uniqueness.

MM7033-M1.02 · E · 15 min
Learning objective: U5 Tensor computation
Compute $\mathbb{Z}/15\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/21\mathbb{Z}$.
The tensor is $\mathbb{Z}/(15\mathbb{Z} + 21\mathbb{Z}) = \mathbb{Z}/3\mathbb{Z}$, by the cyclic tensor formula and Bézout.
Check your argument. Name the canonical generator tensor.

MM7033-M1.03 · C · 30 min
Learning objective: U3–U4 Presentations
Identify the module generated by a, b with relations $a = 3b$ and $10b = 0$, and decide whether it is projective over $\mathbb{Z}$.
The module is $\mathbb{Z}/10\mathbb{Z}$ via $b \mapsto [1]$, $a \mapsto [3]$, with the inverse sending $[1]$ to b. It is not projective: lifting its identity across the quotient map would give a section, but the image of $[1]$ would be an integer killed by ten and hence zero, contradicting the section equation.
Check your argument. Give the isomorphism and a separate projectivity argument.

MM7033-M1.04 · C · 30 min
Learning objective: U6 Localization
Compute $\mathbb{Z}\left[\frac{1}{3}\right] \otimes_{\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$.
The tensor is $\mathbb{Z}\left[\frac{1}{3}\right]/12\mathbb{Z}\left[\frac{1}{3}\right] = \mathbb{Z}\left[\frac{1}{3}\right]/4\mathbb{Z}\left[\frac{1}{3}\right]$. Since three is already invertible modulo four, this quotient is $\mathbb{Z}/4\mathbb{Z}$. The natural reduction map sends $\frac{a}{3^n}$ to $[a][3]^{-n}$.
Check your argument. Explain the last localization identification.

MM7033-M1.05 · A · 40 min
Learning objective: U2–U3 Simple quotients
Prove every nonzero finitely generated module over a commutative unital ring has a maximal proper submodule.
The poset of proper submodules is nonempty. For a nonempty chain, its union is a submodule; the empty chain has the proper upper bound zero because M is nonzero. If the union were all of M, finitely many generators would all lie in one member, forcing that member to be M, a contradiction. Thus every chain has a proper upper bound. Zorn gives a maximal proper submodule, whose quotient is simple.
Check your argument. Use finite generation exactly in the properness check.

MM7033-M1.06 · A · 40 min
Learning objective: U7–U8 Operators
Let $T^3 = 1$ over $\mathbb{C}$. Prove the space is a direct sum of the three eigenspaces and describe projections using interpolation.
For roots λ_i, set $e_{i(t)} = \prod_{j \neq i} (t - \lambda_j)/(\lambda_i - \lambda_j)$. Modulo $t^3 - 1$, these are orthogonal idempotents summing to one and satisfy $(t - \lambda_i)e_i = 0$. Substituting T gives projections onto the corresponding eigenspaces, with sum the identity and pairwise zero products. Their images therefore give the direct-sum decomposition.
Check your argument. Explain why polynomial congruences remain true after substituting T.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M2.01 · E · 15 min
Learning objective: U3 Module maps
Compute $\mathrm{Hom}_{\mathbb{Z}(\mathbb{Z}/6\mathbb{Z},\mathbb{Z}/9\mathbb{Z})}$ as an abelian group.
The allowable images are $[0]$, $[3]$, $[6]$ modulo nine. Under pointwise addition these form a cyclic group of order three, so the Hom group is $\mathbb{Z}/3\mathbb{Z}$.
Check your argument. Identify the group operation as well as the number of maps.

MM7033-M2.02 · E · 15 min
Learning objective: U7 Exterior powers
Compute the rank of $\wedge^3 R^5$ and explain why $\wedge^6 R^5 = 0$.
The first rank is $\binom{5}{3} = 10$. In six slots drawn from five basis indices, some index repeats, so every basis expansion term vanishes and the sixth exterior power is zero.
Check your argument. Justify both spanning and the repeated-index argument.

MM7033-M2.03 · C · 30 min
Learning objective: U1 Splitting fields
Find the degree of the splitting field of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.
The splitting field is $\mathbb{Q}\big(\sqrt{2},\sqrt{3}\big)$. If $\sqrt{3} = a + b\sqrt{2}$, squaring forces $ab = 0$, and either $a^2 = 3$ or $b^2 = \frac{3}{2}$, impossible for rational a, b. Thus each adjoining step has degree two and the total degree is four.
Check your argument. Prove the relative second degree, not merely an upper bound.

MM7033-M2.04 · C · 30 min
Learning objective: U8 Noetherianity
Prove $K[x,y]/(x^2, xy)$ is Noetherian. Must it be a domain?
Hilbert's theorem makes $K[x,y]$ Noetherian, and its quotient is Noetherian. It is not a domain: the class of x is nonzero because the ideal is generated by degree-two monomials, but its square is zero.
Check your argument. Separate Noetherianity from absence of zero divisors.

MM7033-M2.05 · A · 40 min
Learning objective: U5–U6 Local detection of a map
Prove a module map is an isomorphism if its localization at every maximal ideal is an isomorphism.
Exactness identifies localized kernels and cokernels with the kernels and cokernels of the localized maps, so both localize to zero at every maximal ideal. The local zero-detection theorem forces the original kernel and cokernel to be zero. The map is therefore injective and surjective, hence an isomorphism.
Check your argument. No finite-generation assumption is needed; cite the element-annihilator proof of zero detection.

MM7033-M2.06 · A · 40 min
Learning objective: U4–U7 Projective trace
Let M be finite projective and $T(m) = \varphi(m)v$. Prove $\mathrm{tr}(T) = \varphi(v)$ using a dual basis.
For a dual basis (m_i, ψ_i), $\mathrm{tr}(T) = \sum_i \psi_{i(\varphi(m_i)v)} = \sum_i \varphi(m_i)\psi_{i(v)}$. Since $v = \sum_i \psi_{i(v)} m_i$, applying φ gives exactly this sum. Thus the trace is $\varphi(v)$, without assuming a global ordinary basis.
Check your argument. Use scalar commutativity and the correct dual-basis identity.